**Enhanced Caesar Cipher:**

For part 2 of our project, our group decided to tackle the modified Caesar cipher document and add a layer of complexity to it, hardening the encryption. We did this by implementing dynamic substitution and scrambling.

**Dynamic Substitution:**

Our dynamic substitution uses tables (Upper, Lower, Numerical) to determine a character's position, and then shifting its value within its respective table based on its index “p”. If the index is even, the system shifts the character right by the value (p + key). If the index is odd, the system shifts left (p - key), with key being a variable chosen by the users.

We decided to go with a variable key for this implementation because it adds a large amount of resistance against brute-force attacks, with only a very minor tweak of the original implementation, which shifts by either (p + 1) or (p - 1).

**Positional Scrambling:**

Our positional scrambling implementation uses a columnar transposition layer. We create the grid for the transposition by using (key mod 10) (Remember that our key varies based on user choice, so grid size may vary). As our plaintext is passed through our grid, we read it column-by-column instead of row-by-row, changing the encrypted text immensely, making frequency and brute force

attacks much harder.

# EXAMPLES:

1. Plaintext="crypto" with key=5

↳ c: p=2 (even) → p+5 = 7 = h
r: p=17 (odd) → p-5 = 12 = m
y: p=24 (even) → p+5 = 29 mod 26 = 3 = d
p: p=15 (odd) → p-5 = 10 = k
t: p=19 (odd) → p-5 = 14 = o
o: p=14 (even) → p+5 = 19 = t

} hmdkot

hmdkot:

| 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
| h | m | d | k | o |
| t | | | | |

Ciphertext = htmdko

2. Plaintext="Jhoan" with key=3

↳ J: p=9 (odd) → p-3 = 6 = G
h: p=7 (odd) → p-3 = 4 = e
o: p=14 (even) → p+3 = 17 = r
a: p=0 (even) → p+3 = 3 = d
n: p=13 (odd) → p-3 = 10 = k

} Gerdk

Gerdk:

| 1 | 2 | 3 |
|---|---|---|
| G | e | r |
| d | k | |

Ciphertext = Gdekr